Math 407 Section 1
Fall 2025
Exam 3
April 16, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): ____________________________

Student ID: ____________________________

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- If the problem asks for a proof, be sure to carefully justify your work, including any theorems from class.

  Note: you may NOT use a theorem or result from class to prove something when it makes the problem entirely trivial. If you are unsure whether a particular theorem or result is allowed, just ask!

  Unless otherwise stated, all rings will be commutative with identity and all ring homomorphisms will send the identity to the identity.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

   (a) Every zero divisor is nilpotent

   (b) Any unique factorization domain is a principal ideal domain

   (c) Every field is a subfield of the complex numbers

   (d) In a principal ideal domain, any two elements have a greatest common divisor

   (e) The element $7 + \sqrt{5}$ is algebraic over $\mathbb{Q}$

2. (10 points)

   (a) State the definition of prime ideal of a ring R

   (b) State the definition of a prime element of R

   (c) Prove that $a \in R$ is prime if and only if (a) is a prime ideal

3. (10 points)

   (a) Write the definition of a maximal ideal.

   (b) Suppose that $f : R \to S$ is a ring homomorphism. Prove that

   $$\ker(f) = \{r \in R : f(r) = 0_S\}$$

   is an ideal of the ring R.

   (c) Prove that if R is a field in (b), then either f is trivial or f is injective.

4. (10 points)

Consider the polynomials ring $\mathbb{Z}[x]$ of polynomials with integer coefficients, the polynomial $p(x) = x^2 + x + 1$, and the ideal

$$I = (2, p(x)) = \{2f(x) + p(x)g(x) : f(x), g(x) \in \mathbb{Z}[x]\}.$$

(a) Show that any polynomial $f(x) \in \mathbb{Z}[x]$ is equivalent to a polynomial of degree at most 1 modulo I.

(b) Show that any polynomial $f(x) \in \mathbb{Z}[x]$ is equivalent to 0, 1, x, or $x + 1$ modulo I.

5. (10 points)

   Carefully prove that
$$\mathbb{Q}(\sqrt{3}, \sqrt{5}) = \mathbb{Q}(\sqrt{3} + \sqrt{5})$$
.

6. (10 points) Let K be a field extension of F and suppose that $u \in K$.

   (a) Define the minimal polynomial of an element $u \in K$ which is algebraic over F.

   (b) Prove that if $f(x)$ is the minimal polynomial of u, then $f(x)$ is irreducible.

   (c) Prove that if $u \in K$ and $[K : F] = n$, then the minimal polynomial of u has degree at most n.

7. (10 points) Let $F = \mathbb{Q}$, $K = \mathbb{Q}(\sqrt[3]{5})$ and $E = \mathbb{Q}(\sqrt[3]{5}, \sqrt{7})$.

   (a) Find a basis for K as an F-vector space. Justify your answer!

   (b) Find a basis for E over K. Justify! You may use without proof the fact that $\sqrt{7} \notin K$.

   (c) Find a basis for E over F. Justify!